4. Synthesis, Contradictions, Blind Spots, Research Gap

4.1 Three points of convergence

Internal validation isn't sufficient evidence of generalisation. The case for this being genuine consensus is well made - four papers from four different traditions all arriving at the same conclusion. Christodoulou gives the numbers (0.72 → 0.54), Sullivan gives the theoretical explanation (conflating statistical generalisation with practical robustness), Whalen gives the methodological mechanism (leaky preprocessing inflates internal performance), Collins responds normatively (makes external validation mandatory). That convergence across empirical, theoretical, methodological, and normative traditions does feel like something close to settled.

Dataset shift is everywhere and poorly handled. The diagnosis is consistent - Wörheide in clinical biomarkers, Springer and Doering in immunological sequence data, Ben-David mathematically, Subbaswamy and Saria pointing toward causal mechanisms as the only durable solution. The important observation here is that the diagnosis is consistent but the solution isn't. The field agrees on what the problem is and hasn't converged on what to do about it. That asymmetry matters.

Spurious correlations aren't the same as sampling noise. Emerson's simulations show models learning confounded relationships that collapse at deployment. Springer and Doering's decoy experiment shows models memorising receptor sequence features while appearing to learn interaction patterns. Whalen identifies confounding as a pitfall. All three agree statistical corrections alone aren't enough - need to get at the mechanism. The unanswered question is how.

4.2 Two genuine contradictions

Statistical vs. causal framing of generalisation failure. This is probably the most important unresolved disagreement in the whole literature. Statistical tradition (Whalen, Collins, Christodoulou): failure is a problem of discipline and data quality - get the splits right, report honestly, collect more diverse data. Causal tradition (Emerson, Arjovsky, Subbaswamy and Saria): even with perfect discipline and data handling, a model learning statistical associations will learn the wrong ones when those associations are context-dependent. Both positions are empirically supported. Neither has been tested against the other.

The specific experimental gap is worth noting precisely: Christodoulou tests naive standard methods and finds they fail, but never tries anything from the causal toolkit. Emerson tests causal modelling and shows it helps, but doesn't compare against a carefully implemented importance-weighting approach. The result is that nobody knows - empirically - under what conditions better statistical practice solves the problem vs. when causal machinery is actually required. That experiment is absent from the literature.

Applied optimism vs. theoretical caution on achievability. Wörheide and Emerson are broadly optimistic - shift can be corrected, robustness can be improved. Ben-David is more cautious at the theoretical level - when divergence between source and target distributions is large enough, no correction strategy can recover good performance because the distributions are simply too different. Applied papers don't engage with this theoretical bound. The theory paper doesn't test its implications on real clinical data. The field therefore lacks a calibrated sense of when correcting for shift is worth attempting vs. when the honest answer is "this model cannot work here regardless." That's critical information for deployment decisions that currently doesn't exist.

4.3 Three blind spots

Missing middle ground between detecting failure and preventing it. The literature is good at documenting failures retrospectively and at proposing theoretical solutions. What's almost entirely absent is the prospective question: can the expected performance gap be estimated before deployment? Can it be predicted whether a model will generalise to a specific new context before committing to that deployment? None of the fifteen papers surveyed provides a validated framework for that prospective assessment. They can tell you the model failed. They can't tell you in advance that it was likely to.

Monitoring. A deployed clinical prediction model doesn't stay static - patient populations change, treatment protocols evolve, measurement technologies improve, coding practices drift. The model that worked at deployment may be quietly degrading. Collins acknowledges this in Item 26, Wörheide mentions continuous monitoring as desirable. But no paper treats monitoring as a primary research question. What signals indicate drift? How do you distinguish genuine shift from normal sampling variation? When do you trigger recalibration? What does recalibration look like in a live clinical system? These aren't minor implementation details - they're fundamental to whether deployed models remain safe and useful over time.

The DAG specification problem. Emerson's causal framework is the most constructive contribution in the corpus and the direction it points seems right. But it requires specifying a causal graph before the framework can help. In most clinical ML settings, that prior knowledge doesn't exist in reliable complete form. Relationships are contested, partially understood, domain-specific, and requiring expertise most ML teams don't have. The gap between "causal methods are the right approach" and "here is how to apply them when there's no validated causal graph" is the most significant unresolved methodological problem in the whole literature. The paper identifies this gap clearly and it's the most direct pointer toward what original research could contribute.

4.4 The research gap

The gap statement is built from accumulation rather than assertion, which is methodologically honest. The literature has: established that standard models fail at deployment (Christodoulou); characterised dataset shift mechanisms and correction strategies for known shifts (Wörheide, Ben-David); proposed invariant and causal learning as principled alternatives (Arjovsky, Emerson); mapped the extreme boundary condition where extrapolation to genuinely novel categories fails completely (Springer and Doering); defined minimum reporting standards (Collins).

What's missing is a framework that does four things together:

1. Quantify - before deployment - the expected generalisation gap between training population and target deployment population
2. Identify from data which features are likely to remain predictive across anticipated shift and which are shift-sensitive and unreliable
3. Give practitioners actionable guidance when the gap is too large - adapt, restrict, or don't use
4. Integrate with ongoing post-deployment monitoring that detects distributional drift before performance has already degraded

The theoretical components exist, scattered across five clusters. What doesn't exist is an integrated, clinically validated, operationally deployable version that doesn't require a fully specified causal graph or multiple independent training environments - resources almost never available in practice.

The final characterisation of the gap is worth noting: explicitly framed as not a gap in awareness. The field is diagnostically sophisticated. The problem is that no one has assembled the pieces into something that works end-to-end in realistic conditions without heroic prior knowledge or data requirements. That framing is important because it defines exactly what kind of contribution would be original - not identifying a new problem, but solving a known assembly problem under realistic constraints.

Cross-section observations

The section's strongest move is the precise characterisation of the prospective vs. retrospective asymmetry - the literature can tell you a model failed, but can't tell you in advance that it was likely to. That specific framing of the gap is more actionable than a generic "more research needed" conclusion.

The two unresolved contradictions (statistical vs. causal framing; applied optimism vs. theoretical caution) are the most intellectually interesting parts of the section. Both point toward specific experiments that haven't been done rather than just knowledge that's missing. That's a more useful kind of gap to identify.

The monitoring blind spot feels underweighted given how much emphasis is placed on the prospective assessment problem. Arguably detecting drift post-deployment is just as important as predicting it pre-deployment, and the literature's silence on it is striking given how long clinical ML systems have been deployed in practice.